

NASA CR-114614

V/STOL Tilt Rotor Study - Volume V

A Mathematical Model for Real-Time Flight Simulation of the Bell Model 301 Tilt Rotor Research Aircraft

SOURCE STATUS

The original NASA PDF download endpoint currently returns HTTP 404 to ordinary download clients. This file is a source-verified technical extract reconstructed from the NASA NTRS indexed full text and citation record. It is not a facsimile copy and does not preserve every figure, equation layout, or scan artifact. Use it for model-structure and parameter-traceability work; do not use it for graphical digitization.

Report number	NASA-CR-114614; Bell 301-099-001, Volume V
Publication date	13 April 1973
Authors	P. B. Harendra; M. J. Joglekar; T. M. Gaffey; R. L. Marr
Contract	NAS2-6599
NASA document ID	19730022217
NTRS citation	https://ntrs.nasa.gov/citations/19730022217
Original PDF name	19730022217.pdf
Indexed extent	268 PDF pages in the NTRS indexed copy
Distribution	Public; NTRS citation states public use permitted

Purpose of this extract

This extract preserves the parts needed for the current full-angle wing-model workflow: the separation of freestream and rotor-wake wing loads, wake-immersed area, local flow quantities, left/right asymmetry, high-negative-angle aerodynamic data, and the limits of the original Model 301 empirical representation.

1. Model architecture relevant to the new wing model

Separate load paths

The report computes aerodynamic forces and moments caused by the rotor wake separately from the forces and moments caused by freestream flow. This is the key architectural point: wake and freestream regions are spatial regions with different local flow, not two complete aerodynamic laws blended by flight speed.

Wake-immersed wing region

For the portion of the wing inside a rotor wake, the model determines the immersed area, local angle of attack, and local dynamic pressure before computing lift and drag.

Freestream remainder

The wing area outside the wake is evaluated with the freestream local-flow quantities. The two area contributions are then combined as physical area integrals.

Large negative incidence

At low speed, rotor wake can drive the wing to large negative incidence. The report states that lift and drag tables are defined to approximately 90 degrees in magnitude for this regime.

Asymmetric operation

Left and right wakes are treated separately. Unequal wake coverage during asymmetric flight naturally produces roll and yaw moments.

Project implication

The replacement wing model should use one common full-angle coefficient database for every strip. Wake and freestream portions may differ through local velocity, angle of attack, Reynolds number, Mach number, dynamic pressure, covered area, and moment arm. They should not be represented by a speed-dependent blend of complete FNear and FLiftLine force/moment results.

2. Minimum mathematical form

For each left/right wake and each wing strip, define the wake-covered area S_{iW} and local flow state. A modern implementation consistent with the report structure can be written as:

$$q_{iW} = 0.5 \rho |V_{iW}|^2$$

$$L_{iW} = q_{iW} S_{iW} C_L(\alpha_{iW}, Re_{iW}, M_{iW}, \delta)$$

$$D_{iW} = q_{iW} S_{iW} C_D(\alpha_{iW}, Re_{iW}, M_{iW}, \delta)$$

$$M_{c/4,iW} = q_{iW} S_{iW} c C_m(\alpha_{iW}, Re_{iW}, M_{iW}, \delta)$$

The freestream part uses $S_{free} = S_{strip} - S_{iW}$ and the same coefficient law evaluated at the freestream local state. Total strip force and moment are the sum of the physical area contributions.

These equations are a clean modern restatement of the report architecture. They are not claimed to reproduce every symbol or empirical constant in the 1973 implementation.

3. Wake coverage variables identified by CR-114614

Quantity	Role in the original representation	Recommended project treatment
Wake radius	Defines the cross-section of the contracted wake at the wing.	Parameterize wake radius at the wing plane; derive from rotor radius, contraction model, and disk-to-wing distance.
Conversion / mast angle	Changes rotor-axis direction and therefore the wake centerline intersection with the wing.	Use the actual rotor-axis unit vector and intersect the wake tube with the wing-strip geometry.
Wake angle of attack	Changes the orientation of the wake relative to the wing.	Compute from local velocity vector, not a model-switch weight.
Fuselage sideslip	Changes left/right immersed areas and asymmetric loads.	Treat left and right wakes and half-wings independently.
Vertical distance below disk	The report notes wing and stabilizer surfaces are about 0.4R below the rotor disk.	Keep distance parameterized; use source-traceable geometry in validation cases.
Induced velocity	Varies with speed and mast tilt; original constants K_i came from powered-model data.	Prefer the current rotor model output. Do not copy Model 301-specific K_i without evidence.

The report states that contracted wake stability was observed down to the wing and horizontal-stabilizer surfaces and simplified the hover wake-radius relation because these surfaces were located at approximately 0.4 rotor radius below the disk.

4. Source limitations that must remain visible

- The 1973 model was constrained by real-time computing limits and used simplified rotor aerodynamics and uniform inflow.
- The rotor-wake/airframe interaction model relied on limited powered-model data.
- The induced-velocity constants were specific empirical fits, not universal tiltrotor constants.
- The report itself recommends extending airframe aerodynamics to plus/minus 180 degrees and improving above-stall downwash representation.
- A modern full-angle wing database should therefore use CR-114614 for structural guidance and geometry definitions, while using XFOIL, TM-88373, and explicit post-stall extensions for the coefficient data chain.

5. Traceability map for the current workflow

Workflow item	Primary source	Status enabled by this extract
Separate wake and freestream area loads	CR-114614 Section II.D and Appendix A	Direct structural evidence
Wake area variables	CR-114614 Section II.D / Figure II-15 description	Direct variable list
Disk-to-wing scale near 0.4R	CR-114614 Section II.D	Direct geometric scale
Left/right asymmetry	CR-114614 Section II.D	Direct physical requirement
High-negative-angle tables to near 90 deg	CR-114614 Section II.D	Direct range statement
Near-vertical CL/CD/Cm curves	NASA TM-88373	Use original PDF in this bundle
Rotor/wing download strip validation	NASA CR-176970	Use original PDF in this bundle
NACA 6/6A ordinate generation	NASA TM-X-3069 and TM-4741	Use original PDFs in this bundle
Published 6A validation tables	NACA TR-903	Use original PDF in this bundle

6. Codex use instructions

Treat this document as an authenticated technical extract, not as a graphical facsimile. It is adequate to unblock source tracing and implementation of the wake-strip architecture. For any task that requires reading Figure II-15 geometry, exact scanned equations, or numerical tables not reproduced here, retain the gate as PARTIAL until the original scan becomes downloadable or another authenticated copy is obtained.

Do not restart browser DOM inspection or repeatedly request the broken direct endpoint. Continue all independent work using the verified architecture above and the original TM-88373, CR-176970, TM-X-3069, TM-4741, and TR-903 files in the bundle.

7. Official source reference

NASA Technical Reports Server citation record: <https://ntrs.nasa.gov/citations/19730022217>

Original file name listed by NTRS: 19730022217.pdf. Normal download clients returned HTTP 404 during package construction on 3 July 2026, while the indexed NTRS full text remained available to the search service.